



KALP PATEL

Kalp2801@gmail.com • (408)-581-1584 • www.linkedin.com/in/drkalppatel/ • <https://github.com/Kalp25740>

EDUCATION

Indiana University, Indianapolis	MS in Health Informatics GPA: 3.7	Expected May 2025
Maharashtra University of Health Sciences	Bachelor of Dental Surgery GPA: 3.8	May 2013

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, Seaborn, PyTorch, Tensorflow), R Programming, SAS
Data Analysis Techniques: Natural Language Processing (Spacy, NLTK, Transformers), Machine Learning, Statistical Analysis, Time-Series Regression.
Data Management Tools: SQL (MySQL), Snowflake, GitHub, NVivo, Qualtrics, REDCap, Microsoft Office (Excel, Outlook)
Data Visualization Tools: Tableau, Power BI, R Shiny
Healthcare Standards: CPT, SNOMED, ICD, LOINC, FHIR, HL7, HIPAA

PROFESSIONAL EXPERIENCE

Graduate Research Data Analyst

Purkayastha Lab of Health Innovations, Indiana University

Jan 2024 – Present

- Led data cleaning and preprocessing of 200,000+ patient records, ensuring data quality, temporal alignment, and PHI compliance; assisted with IRB documentation and approvals; performed feature extraction and developed time series predictive models to forecast wound healing trajectories, supporting early identification of high-risk patients for timely clinical intervention.
- Evaluated an AI-powered diagnostic tool in a multi-site study involving 40 clinicians; performed quantitative analysis using Python and R, qualitative analysis with NVivo software, and generated implementation reports to guide clinical decision-making.
- Integrated and preprocessed large-scale datasets from census and state health sources; performed statistical analyses and applied machine learning techniques to identify significant predictors of social determinants of health, translating results into actionable reports and presentations to support public health decision-making.
- Designed and implemented end-to-end data pipelines to merge, clean, and analyze 18M+ clinical records from the MIMIC-IV database; applied statistical modeling and time-series analysis to predict patient outcomes as part of an app aimed at identifying children at high risk of mortality and supporting early clinical intervention.

Informatics Administrative Intern

Clinical Architecture

May 2024 – Aug 2024

- Standardized and integrated healthcare data from disparate EMR systems to ensure interoperability; conducted data analyses on 200+ operational issues and prepared comprehensive reports highlighting trends and actionable insights to enhance customer experience and drive quality management initiatives.

Biostatistics Teaching Assistant

Indiana University Indianapolis

Aug 2024 – Dec 2024

- Mentored 60 students in advanced statistical methods, providing comprehensive guidance on R statistical software through weekly sessions and project support, and developing instructional materials to enhance statistical programming skills.

Senior Dentist

Dental Inn

April 2018 – Oct 2022

- Successfully managed dental practice operations and modernized patient record management by integrating a new Electronic Health Records (EHR) system, resulting in improved data accessibility.
- Performed SQL-based abstraction and review of 1,000+ patient records to identify care trends and optimize appointment scheduling, increasing operational efficiency by 20%.

PUBLICATIONS

The Impact of Social Determinants on Cardiovascular Mortality: A Zip Code-Level Analysis in Indiana – Accepted Full Paper, MedInfo2025, International Medical Informatics Association (IMIA), August 2025

ACADEMIC PROJECTS

Healthcare Diabetes Risk Prediction (Python)

- Developed predictive machine learning models for diabetes risk assessment, achieving 98% accuracy, applying advanced data preprocessing and statistical analysis techniques to identify critical predictive factors for early diagnostic intervention.

Statistical Analysis of Factors Associated with Obesity (R Studio)

- Evaluated 2,000 records using R Studio, applying descriptive statistics, regression modeling, and categorical data analysis to identify significant obesity correlations ($p < 0.05$). Generated comprehensive visualizations to communicate key findings.

Public Health Analytics and Decision Support using DHIS2 (R and DHIS2)

- Developed forecasting models (SARIMA, Holt-Winters) on 24 months of health data to predict maternal mortality and vaccination trends, improving forecasting accuracy by 22% (MAPE); designed interactive dashboards and geospatial maps in DHIS2 to track critical health indicators across 20+ facilities, identifying high-risk regions and informing resource allocation strategies.